

**MAT1164 : Calculus for Engineers**

Date: 23/12/ 2014 Duration: 2 hours, Marks: 50

Instructions:

1. Only calculators can be used. Printed materials and Smart phones are not allowed.
2. There is SECTION A and SECTION B. Answer all questions in Section A. Choose ONLY 2 questions in Section B.
3. Do not forget to write your Registration Number.
4. Start new question from a fresh page.

SECTION A. Answer all the question in this section. (20 marks)**Question 1**(a) Let f be the function defined as

$$f(x) = \begin{cases} 3 - x, & x < 2 \\ 2, & x = 2 \\ \frac{x}{2}, & x > 2 \end{cases}$$

Does the function f continuous at $x = 2$? (4 marks)(b) Let $w = xy + yz + xz$, $x = u + v$, $y = u - v$, and $z = uv$.(i). Express $\partial w / \partial u$ and $\partial w / \partial v$ as functions of u and v using the Chain Rule. (3 marks)(ii). Evaluate $\partial w / \partial u$ and $\partial w / \partial v$ at the given point $(u, v) = (1/2, 1)$. (3 marks)

(c) Evaluate

$$\int_0^{\pi/4} e^t \sin 2t dt$$

(4 marks)

$$y = \frac{xy}{x} \cdot \frac{dx}{du} \cdot \frac{dv}{du}$$

$$x = u + v \\ y = u - v$$

- (d) Find the measures of the angles between of the triangle whose vertices are : $A = (-1, 0)$, $B = (2, 1)$, and $C = (1, -2)$. (5 marks)

SECTION B. Answer any 2 questions in this section (30 marks)

Question 2: Let f be the function defined as follows:

$$f(x) = \frac{3}{4}(x^2 - 1)^{\frac{2}{3}}$$

- (a) What is the domain of f ? (1 mark)
- (b) Identify the intervals on which f is increasing and decreasing. (5 marks)
- (c) Determine the maximum point(s) and minimum point(s) of f if do exist. (4 marks)
- (d) Determine the inflection point(s) of f and identify the intervals on which f is concave up and concave down. (5 marks)

Question 3 ✓

- (a) Solve $z^2(1 - z^2) = 16$. (7.5 marks)
- (b) A rectangular sheet of metal having dimensions 20cm and 12cm has squares removed from each of the four corners and the sides bent upwards to form an open box. Determine the maximum possible volume of the box. (7.5 marks)

Question 4 ✓

- (a) Given the following system of linear equations:

$$\begin{cases} x + 2y + z = 4 \\ 3x - 4y - 2z = 2 \\ 5x + 3y + 5z = -1 \end{cases}$$

- (i) Write the system in matrix form $Ax = b$ (1 mark)

- (ii) Find the vector $x = A^{-1}b$. (7 marks)

- (b) Given the points $P(2, -2, 1)$, $Q(3, -1, 2)$, and $R(3, -1, 1)$

- (i) Find the area of triangle determined by the points P, Q , and R . (3.5 marks)
- (ii) Find the unit vector perpendicular to plane PQR . (3.5 marks)

End